Rajeev Gandhi Memorial College of Engineering & Technology

(Autonomous)

Nandyal – 518501, Nandyal-Dt. AP. India

Department of Computer Science & Engineering (Data Science)

IV B. Tech Project Report A.Y. 2025-26 (Batch-6)

Project & Batch Details

- **Academic Year / Batch:** 2022 - 2026
- **Section:** C
- **Number of Students in the Batch:** 6
- **Project Title:**

  Real-Time Helmet and Camera-Mounted Device Detection with License Plate Recognition Using YOLOv8
- **Student Names & Registration Numbers:**
  1. K. Sravani (22091A32E2)
  2. D. Sujitha (22091A32E8)
  3. M. Umarani (22091A32F7)
- **Project Guide Name:** Dr. M. Suleman Basha

Major Areas of the Project

- **Business Domain:** Intelligent Transportation Systems / Road Safety Automation
- **Research Area Involved:** Computer Vision, Deep Learning, Object Detection, Traffic Rule Violation Detection
- **Technology / Tools Adopted:** Python, YOLOv8, OpenCV, CNN, OCR (ALPR), Labellmg, Numpy, Pandas, CSV Export

Expected Outcomes of the Project

- **Business Outcomes:**

  Improves road safety enforcement using surveillance infrastructure with Automates detection of traffic rule violations such as helmetless riding and camera-mounted helmets.
- **Research Outcomes:**

  Demonstrates effectiveness of YOLOv8 for real-time helmet and number plate detection and validates custom dataset training for real-world traffic scenarios with high accuracy.
- **Application Outcomes:**

  Provides a working automated traffic violation detection system which Generates offender vehicle numbers using OCR and exports them into structured CSV files.

Work Division Details

Clearly mention how the project work is divided:

1. Project Understanding & Scope Definition
2. Study of traffic rule violations and problem identification.
3. Dataset Collection & Image Labelling.
4. Capturing real-world road images and labeling using Labelling.
5. Model Development & Training.
6. YOLOv8 model training for helmet, camera, and no-helmet detection.
7. OCR Integration & Data Processing.
8. Number plate extraction and CSV generation.
9. Testing, Evaluation & Documentation.
10. Performance analysis, accuracy calculation, report preparation.

Industry Expert for Project Evaluation

- **Name of Industry Expert** :
- **Designation** :
- **Organization / Company Name** :
- **Area of Expertise** :
- **Official Email ID** :
- **Contact Number** :
- **Nature of Association** :

Tentative Project Evaluation Schedule

- **Preferred Date(s)** : 1st Week of March, 2026
- **Time Slot** :
- **Mode** (Google Meet / MS Teams / Zoom):
- **Duration of Evaluation** :

Suitable Journals

1. Deep-Learning Based Helmet Violation Detection System
Summary: YOLO-based deep learning model used to detect helmet violations of motorbike riders in real-time traffic surveillance.

https://ieeexplore.ieee.org/document/9670937

IEEE: N. Kharade et al., "Deep-learning based helmet violation detection system," IEEE AIMV, 2021.

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2. Helmet Violation and Number Plate Detection System

Summary: Automated system that detects helmet violations and extracts vehicle number plates using YOLO and OCR techniques.

https://ieeexplore.ieee.org/document/10018664

IEEE: R. L. Vardhan et al., "Helmet violation and number plate detection system," IEEE, 2023.

3. Higher Accuracy YOLOv5 Based Safety Helmet Detection

Summary: Enhanced YOLOv5 architecture improves helmet detection accuracy in complex traffic environments.

https://link.springer.com/chapter/10.1007/978-3-031-07004-5_20

IEEE: Z. Wang et al., "Higher accuracy YOLOv5 based safety helmet detection," Springer, 2022.

4. Vehicle and Plate Detection for Intelligent Transport Systems Using YOLOv8

Summary: YOLOv8 model applied for accurate vehicle and license plate detection in intelligent transportation systems.

https://www.sciencedirect.com/science/article/pii/S1877050923004842

IEEE: M. H. F. Afonso et al., "Vehicle and plate detection for intelligent transport systems," Procedia Computer Science, 2023.

5. Real-Time License Plate Detection for Non-Helmeted Motorcyclists Using YOLO

Summary: Real-time system detects non-helmeted riders and identifies their license plates using YOLO and OCR.

https://www.sciencedirect.com/science/article/pii/S2405959520301647

IEEE: Y. Jamtsho et al., "Real-time license plate detection for non-helmeted motorcyclists using YOLO," ICT Express, 2020.

6. Real-Time Multi-Class Helmet Violation Detection Using YOLOv8

Summary: Multi-class YOLOv8 framework detects helmet usage violations in real-time traffic videos.

https://ieeexplore.ieee.org/document/10203971

IEEE: A. Armstrong et al., "Real-time multi-class helmet violation detection using YOLOv8," IEEE CVPRW, 2023.

7. Automated Non-Helmet Rider Detection Using YOLO and OCR

Summary: YOLO and OCR-based system automatically identifies helmetless riders and logs their vehicle numbers.

https://ieeexplore.ieee.org/document/10353406

IEEE: S. Prabhakaran et al., "Automated non-helmet rider detection using YOLO and OCR," IEEE GCAT, 2023.

8. YOLOv8-Powered Multi-Object Detection for Safety Helmets and Number Plates in Real-Time Surveillance Systems

Summary: Real-time multi-object detection using YOLOv8 for helmet and number plate identification in traffic surveillance.

https://ieeexplore.ieee.org/document/10895606

IEEE Xplore.

9. Real-Time Vehicle Classification and License Plate Recognition via Deformable Convolution-Based YOLOv8 Network

Summary: A deformable convolution-enhanced YOLOv8 model for real-time vehicle classification and license plate recognition with high accuracy.

https://doi.org/10.1109/JSEN.2024.3453498

10. AI-Powered Traffic Surveillance: License Plate Recognition with Non-Helmet Detection Using YOLOv8

Summary: Traffic surveillance system using YOLOv8 for license plate recognition and non-helmet detection in real-time.
https://ieeexplore.ieee.org/document/10779944

Signature of Supervisor: **Signature of HOD:**

(Dr. M. Suleman) **(Dr. B. Bhaskara Rao)**